Math Inquirers

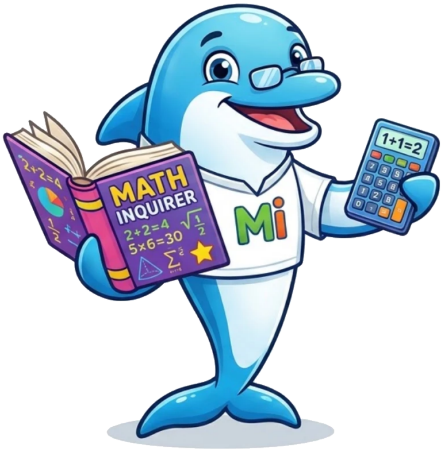
Think. Question. Prove.

Level 1 (ages 6-8) - Term 1

Home Practice Pack

Book 1

Lessons 1 to 5



One sheet for each lesson. Tear out a sheet and do it at home.

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Bundles and Singles: practice at home

Name _____ Date _____

About 15 minutes, a few days after the lesson. Use real things or draw it. Go further (question 5) is optional. Small things stay on the table, away from mouths and little brothers and sisters. A grown-up may read the questions aloud. Wrong answers are fine: they show what to try next.

No.1

Start here

Mei has 3 bundles of ten pencils and 8 single pencils. How many pencils does she have?

No.2

Start here

Fill in: 65 = _____ tens and _____ ones.

No.3

Start here

Write the number: 9 ones and 2 tens.

No.4

Think harder

Ben has 4 tens and 12 ones. Which is his number? Circle one. Then show how you know.

52 412

No.5

Go further

Find 30 to 50 small things. Tie bundles of ten with hair ties or put them in cups. Write: _____ tens and _____ ones = _____. Now open one bundle. Write the tens and ones again.

Answers for grown-ups

Please let your child try first, then check together. Ask *how do you know?* If your child is stuck, wait 10 seconds, then suggest drawing it.

1. 38
3 tens is 30, and 8 more is 38.

2. 6 tens and 5 ones
The left digit is the tens, the right digit is the ones.

3. 29
2 tens is 20, and 9 ones makes 29. The tens digit goes first.

4. 52 (12 ones make 1 ten and 2 ones, so 5 tens and 2 ones)
Tie 10 of the 12 ones into a bundle: 5 bundles and 2 ones, which is 52.

5. Any: for example 3 tens and 7 ones = 37, and also 2 tens and 17 ones = 37. Check: tens x 10 + ones = how many things.
A grown-up counts the things one by one to check the number. Opening a bundle gives one fewer ten and ten more ones.

More practice online (no timers, no scores): ask your tutor for the practice link.

Number Words and the Number Line: practice at home

Name _____ Date _____

About 15 minutes, a few days after the lesson. Use your number cards, a line you draw, or a toy frog. Go further (question 5) is optional. Small things stay on the table, away from mouths and little brothers and sisters. A grown-up may read the questions aloud. Wrong answers are fine: they show what to try next.

No.1

Start here

Fill in the missing numbers: 46 | 47 | _____ | 49 | _____

No.2

Start here

Write the numbers in digits: thirteen, thirty, nineteen, ninety.

No.3

Start here

Fill in the blanks. 83 is between two tens: _____ and _____.

No.4

Think harder

Is 56 nearer to 50 or to 60? Circle one. Then show how you know.

50 60

No.5

Go further

Draw 5 boxes in a row. The jumps are equal. Write numbers in 2 boxes that are not next to each other. Leave 3 boxes empty. Ask a grown-up to solve it. They write the answer on the back.

Answers for grown-ups

Please let your child try first, then check together. Ask *how do you know?* If your child is stuck, wait 10 seconds, then suggest drawing it.

1. 48 and 50
Count on: 47, 48, 49, 50.

2. 13, 30, 19, 90
Teen means ten and some more: 13 and 19. Ty means tens: 30 and 90.

3. 80 and 90
83 is 8 tens and 3 more, so it is past 80 and not yet 90.

4. 60 (4 hops to 60, 6 hops to 50)
56 to 60 is 4 hops. 56 to 50 is 6 hops. 4 is fewer, so 60 is nearer.

5. Any correct puzzle, for example 20 | ____ | ____ | ____ | 40 (jump 5: 25, 30, 35).
A grown-up checks that the jump is the same between every two boxes, that the jump is a whole number, and that only one number fits each empty box.

More practice online (no timers, no scores): ask your tutor for the practice link.

Digit Cards: practice at home

Name _____ Date _____

About 15 minutes, a few days after the lesson. Use your digit cards or draw bars and squares. Go further (question 5) is optional. Small things stay on the table, away from mouths and little brothers and sisters. A grown-up may read the questions aloud. Wrong answers are fine: they show what to try next.

No.1

Start here

You have the digit cards 4 and 7. Write the two numbers you can make. Circle the greater number.

No.2

Start here

Arrange these in order. Begin with the smallest number: 72, 27, 70, 17.

No.3

Start here

Ben says 38 is greater than 41. Is Ben right? Circle Yes or No. I know because _____.

Yes No

No.4

Think harder

You have the cards 0, 6 and 9. Use two cards each time. The smallest two-digit number is _____. The greatest is _____.

No.5

Go further

At home: make digit cards 1, 2, 3, 4 and 5. How many different two-digit numbers can you make with two different cards? Arrange them in order. Begin with the smallest.

Answers for grown-ups

Please let your child try first, then check together. Ask *how do you know?* If your child is stuck, wait 10 seconds, then suggest drawing it.

1. 47 and 74. 74 is greater.
With 4 on the left the number is 47. With 7 on the left the number is 74. 7 tens is more than 4 tens, so 74 is greater.

2. 17, 27, 70, 72
Look at the tens first: 17 has 1 ten, 27 has 2 tens, 70 and 72 have 7 tens. 70 has no ones, so it comes before 72.

3. No. 41 is greater, because 4 tens is more than 3 tens.
41 has 4 tens and 38 has 3 tens. Look at the tens first. The ones digit only decides when the tens are the same.

4. 60 and 96
The 0 can not go on the left. The numbers are 60, 69, 90 and 96. The smallest is 60 and the greatest is 96.

5. 20 numbers: 12, 13, 14, 15, 21, 23, 24, 25, 31, 32, 34, 35, 41, 42, 43, 45, 51, 52, 53, 54
5 choices for the tens card and 4 choices left for the ones card: 5 x 4 = 20. List them with 1 on the left, then 2, then 3, then 4, then 5.

More practice online (no timers, no scores): ask your tutor for the practice link.

The 100 Square Explorer: practice at home

Name _____ Date _____

About 15 minutes, a few days after the lesson. Use your 100 square or draw it. Go further (question 5) is optional. Small things stay on the table, away from mouths and little brothers and sisters. A grown-up may read the questions aloud. Wrong answers are fine: they show what to try next.

No.1

Start here

Count in 5s. Write the next three numbers: 40, 45, 50, _____, _____, _____.

No.2

Start here

On the 100 square a counter is on 58. It moves down 1 row. Which number is it on?

No.3

Start here

(a) Count in 2s: 2, 4, 6, 8, _____ (b) If you keep going, will you say 17? Circle Yes or No.

Yes No

No.4

Think harder

What is the next number? 6, 8, 11, 15, 20, _____.

No.5

Go further

Find a calendar. Pick 3 dates that have another date straight below them. (a) How much more is the date straight below each one? (b) Is it always the same? Write why.

Answers for grown-ups

Please let your child try first, then check together. Ask *how do you know?* If your child is stuck, wait 10 seconds, then suggest drawing it.

1. 55, 60, 65
Each number is 5 more: 50, 55, 60, 65.

2. 68
Down one row is 10 more. 58 and 10 more is 68.

3. 10. No: 17 ends in 7, and counting in 2s ends in 0, 2, 4, 6 or 8.
After 8 comes 10. Numbers counted in twos end in 0, 2, 4, 6 or 8, so 17 is not one of them.

4. 26 (the gaps are 2, 3, 4, 5, then 6)
The gaps are 2, 3, 4, 5. Each gap is 1 bigger, so the next gap is 6 and 20 + 6 = 26.

5. 7 more each time. A row on a calendar is one week of 7 days, like a row of ten on the 100 square.
A calendar row has 7 days, so the date straight below is 7 more. On the 100 square a row has 10 numbers, so the number below is 10 more.

More practice online (no timers, no scores): ask your tutor for the practice link.

Make-a-Ten Adding: practice at home

Name _____ Date _____

About 15 minutes, a few days after the lesson. Use counters or draw a ten-frame. Go further (question 5) is optional. Small things stay on the table, away from mouths and little brothers and sisters. A grown-up may read the questions aloud. Wrong answers are fine: they show what to try next.

No.1

Start here

Fill in the blanks. 9 + 5 = 10 + _____. So 9 + 5 = _____.

No.2

Start here

Add. 37 + 6 = _____.

No.3

Start here

Fill in the missing numbers to make the next ten. (a) 8 + _____ = 10 (b) 45 + _____ = 50 (c) 67 + _____ = 70

No.4

Think harder

Add 5 + 7 + 5 + 3. Show your working. The total is _____.

No.5

Go further

(a) Write 3 sums like 8 + 5 for a grown-up to solve. Use numbers up to 9, like 9 + 6 or 7 + 4. (b) Solve 3 sums that your grown-up writes for you. Many answers are right.

Answers for grown-ups

Please let your child try first, then check together. Ask *how do you know?* If your child is stuck, wait 10 seconds, then suggest drawing it.

1. 4, so 9 + 5 = 14
Move 1 counter from the 5 to the 9. That makes 10, and 4 are left. 10 + 4 = 14.

2. 43
37 needs 3 more to make 40. 6 is 3 and 3. 40 + 3 = 43.

3. 2, 5, 3
Look at the ones digit. 8 needs 2, 5 needs 5, 7 needs 3.

4. 20 (5 + 5 = 10, 7 + 3 = 10)
5 + 5 = 10 and 7 + 3 = 10. 10 + 10 = 20.

5. Any: for example 8 + 5 = 13 (fill the 8 with 2 from the 5: 10 and 3). A grown-up can check with counters.
The number that is nearest to ten is the one to fill. Move the missing amount from the other number, then add what is left.

More practice online (no timers, no scores): ask your tutor for the practice link.